KRAVIONA TECH SOLUTIONS • OFFICIAL PLATFORM GUIDE

Developer & Operator Quick Start Guide

Rapid Onboarding, 5-Step Setup, Test Personas, First Course Authoring, and Common Fixes

Document ID:	DOC-QS-2026-v1.2
Platform Version:	v1.2.0 (Production Release)
Publisher:	Kraviona Tech Solutions (Private Limited)
Release Date:	October 19, 2026
Classification:	Verified Production Technical Manual

CodeMentors LMS — Developer & Operator Quick Start Guide

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1. System Prerequisites & Environment

Before deploying or running CodeMentors locally, verify the following prerequisites:

Requirement	Supported Version	Verification Command	Notes
Node.js	`≥ 20.x` (`24.x` recommended)	`node -v`	Active LTS or Current
NPM	`≥ 10.x`	`npm -v`	Node Package Manager
MongoDB	`≥ 6.x`	`mongod --version`	Local or MongoDB Atlas URI
Web Browser	Chrome `≥ 120` or Edge	`chrome --version`	For UI testing & PDF generation

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2. 5-Step Rapid Setup

Step 1: Clone Repository & Install Dependencies

```
git clone https://github.com/kraviona/codementors.git
cd codementors
npm install
```

Step 2: Configure Local Environment Variables

Create a `.env.local` file in the project root:

```
# Database Connection
MONGODB_URI="mongodb://localhost:27017/codementors"

# Authentication Security
JWT_SECRET="codementors-secure-production-ready-jwt-secret-key-32-chars-min"
DEV_TEST_PASSWORD="DevOnly123!ChangeMe"

# Razorpay Payment Gateway (Test Sandbox Mode)
RAZORPAY_KEY_ID="rzp_test_codementors_demo"
RAZORPAY_KEY_SECRET="codementors_secret_key_mock_12345"
RAZORPAY_WEBHOOK_SECRET="codementors_webhook_secret_mock_12345"

# Platform Metadata
NEXT_PUBLIC_APP_URL="http://localhost:3000"
```

Step 3: Seed Test Accounts & Development Data

Seed the 3 deterministic development accounts:

```
npm run db:seed:test-accounts
```

(Optional: Seed full demonstration catalog with courses, categories, and quizzes):

```
npm run db:seed
```

Step 4: Start the Development Server

```
npm run dev
```

Open blank" rel="noopener noreferrer" class="text-indigo-400 hover:text-indigo-300 underline underline-offset-2">http://localhost:3000 in your web browser.

Step 5: Run Verification Test Suite

Ensure all integration checks pass:

```
npm run test:accounts
```

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3. Dedicated Development Test Personas

> [!WARNING]
> **DEVELOPMENT ONLY:** These credentials are configured strictly for local sandbox testing. Never deploy these credentials to production environments.

CodeMentors LMS — Developer & Operator Quick Start Guide

Document ID: XXINLINECODE0XX
Platform Version: XXINLINECODE1XX
Platform Publisher: Kraviona Tech Solutions
Publication Date: October 19, 2026

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1. System Prerequisites & Environment

Before deploying or running CodeMentors locally, verify the following prerequisites:

Requirement	Supported Version	Verification Command	Notes
Node.js	XXINLINECODE2XX (XXINLINECODE3XX recommended)	XXINLINECODE4XX	Active LTS or Current
NPM	XXINLINECODE5XX	XXINLINECODE6XX	Node Package Manager
MongoDB	XXINLINECODE7XX	XXINLINECODE8XX	Local or MongoDB Atlas URI
Web Browser	Chrome XXINLINECODE9XX or Edge	XXINLINECODE10XX	For UI testing & PDF generation

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2. 5-Step Rapid Setup

Step 1: Clone Repository & Install Dependencies

XXCODEBLOCK0XX

Step 2: Configure Local Environment Variables

Create a XXINLINECODE11XX file in the project root: XXCODEBLOCK1XX

Step 3: Seed Test Accounts & Development Data

Seed the 3 deterministic development accounts: XXCODEBLOCK2XX
(Optional: Seed full demonstration catalog with courses, categories, and quizzes):

XXCODEBLOCK3XX

Step 4: Start the Development Server

XXCODEBLOCK4XX
Open blank" rel="noopener noreferrer" class="text-indigo-400 hover:text-indigo-300 underline underline-offset-2">http://localhost:3000 in your web browser.

Step 5: Run Verification Test Suite

Ensure all integration checks pass: XXCODEBLOCK5XX

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3. Dedicated Development Test Personas

> [!WARNING]
> **DEVELOPMENT ONLY:** These credentials are configured strictly for local sandbox testing. Never deploy these credentials to production environments.

Role	Email Address	Development Password	Portal Route	Primary Function
Administrator	`admin@codementors.test`	`DevOnly123!ChangeMe`	`/admin/dashboard`	Platform Governance, SEO, Users, Payments
Instructor	`instructor@codementors.test`	`DevOnly123!ChangeMe`	`/instructor/dashboard`	Course Authoring, Curriculum, Analytics
Student	`student@codementors.test`	`DevOnly123!ChangeMe`	`/student/dashboard`	Course Player, Quizzes, Certificates

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4. First Course Creation in 3 Minutes

Follow these steps to author your first course:

1. **Sign In:** Navigate to blank" rel="noopener noreferrer" class="text-indigo-400 hover:text-indigo-300 underline underline-offset-2">http://localhost:3000/login and log in as `instructor@codementors.test`.
2. **Access Studio:** Click **"Create Course"** on the dashboard or go to `/instructor/courses/new`.
3. **Basic Information:**

- Title: `Next.js App Router Mastery` - Slug: `nextjs-app-router-mastery` - Category: Select `Web Development` - Skill Level: Select `Intermediate`

1. **Pricing:** Check `Free Foundation Course` (or enter `₹1,999` for paid).
2. **Syllabus:** Author the course description using the **RichTextEditor** with formatting and bullet points.
3. **Save Draft:** Click **"Save Draft"** to generate the course ID.
4. **Build Curriculum:**

- Navigate to `/instructor/courses/[id]/curriculum`. - Click **"Add Module"** (e.g., `Module 1: Foundations`). - Click **"Add Lesson"** inside the module. - Choose **Lesson Type: Video** and provide a stream URL. - Click **"Publish Course"** to deploy the course to the public catalog!

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5. Common Startup Issues & Quick Fixes

Issue	Diagnosis	Quick Resolution
MongoDB connection refused	Local MongoDB daemon is not running.	Run `mongod --dbpath <data_dir>` or provide a valid MongoDB Atlas URI in `MONGODB_URI`.
Redirect loop on /login	Stale or invalid JWT token in cookies.	Clear your browser cookies or open an Incognito window.
Port 3000 already in use	Another development process is occupying port 3000.	Start on a different port: `npx next dev -p 3005` or kill the existing process.
Razorpay modal does not open	Demo keys in production mode.	Toggle gateway mode to `Test` in `/admin/settings/payments` or set valid test keys in `.env.local`.
TypeScript compile errors	Node modules out of sync.	Run `npm run test:rich-text` to verify sanitizer, and ensure Node.js is `≥ 20.x`.

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6. Key Operations Commands Reference

```
# Development
npm run dev                 # Start dev server on port 3000
npm run lint                # Run ESLint validation

# Testing & QA
npm test                    # Run entire test suite (106 tests)
npm run test:accounts       # Verify 3 test accounts & auth
npm run test:rich-text      # Run RichTextEditor & XSS tests
npm run test:instructor     # Run Instructor Studio quota tests

# Production
npm run build               # Compile production Next.js build
npm start                   # Run production server on port 3000
```

Role	Email Address	Development Password	Portal Route	Primary Function
Administrator	XXINLINECODE12XX	XXINLINECODE13XX	XXINLINECODE14XX	Platform Governance, SEO, Users, Payments
Instructor	XXINLINECODE15XX	XXINLINECODE16XX	XXINLINECODE17XX	Course Authoring, Curriculum, Analytics
Student	XXINLINECODE18XX	XXINLINECODE19XX	XXINLINECODE20XX	Course Player, Quizzes, Certificates

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4. First Course Creation in 3 Minutes

Follow these steps to author your first course:

1. **Sign In:** Navigate to blank" rel="noopener noreferrer" class="text-indigo-400 hover:text-indigo-300 underline underline-offset-2">http://localhost:3000/login and log in as XXINLINECODE21XX.
2. **Access Studio:** Click **"Create Course"** on the dashboard or go to XXINLINECODE22XX.
3. **Basic Information:**

- Title: XXINLINECODE23XX - Slug: XXINLINECODE24XX - Category: Select XXINLINECODE25XX - Skill Level: Select XXINLINECODE26XX

1. **Pricing:** Check XXINLINECODE27XX (or enter XXINLINECODE28XX for paid).
2. **Syllabus:** Author the course description using the **RichTextEditor** with formatting and bullet points.
3. **Save Draft:** Click **"Save Draft"** to generate the course ID.
4. **Build Curriculum:**

- Navigate to XXINLINECODE29XX. - Click **"Add Module"** (e.g., XXINLINECODE30XX). - Click **"Add Lesson"** inside the module. - Choose **Lesson Type: Video** and provide a stream URL. - Click **"Publish Course"** to deploy the course to the public catalog!

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5. Common Startup Issues & Quick Fixes

Issue	Diagnosis	Quick Resolution
MongoDB connection refused	Local MongoDB daemon is not running.	Run XXINLINECODE31XX or provide a valid MongoDB Atlas URI in XXINLINECODE32XX.
Redirect loop on /login	Stale or invalid JWT token in cookies.	Clear your browser cookies or open an Incognito window.
Port 3000 already in use	Another development process is occupying port 3000.	Start on a different port: XXINLINECODE33XX or kill the existing process.
Razorpay modal does not open	Demo keys in production mode.	Toggle gateway mode to XXINLINECODE34XX in XXINLINECODE35XX or set valid test keys in XXINLINECODE36XX.
TypeScript compile errors	Node modules out of sync.	Run XXINLINECODE37XX to verify sanitizer, and ensure Node.js is XXINLINECODE38XX.

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6. Key Operations Commands Reference

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